Isaac Marval Hernández

Software Development · Cybersecurity · Applied AI

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PROFILE

Software developer with an Information Systems Engineering background and hands-on cybersecurity experience. Builds applications end to end in Python, Java and SQL, from REST APIs (FastAPI) and databases to full-stack web apps and applied AI systems. On the security side, background in cyber threat intelligence: OSINT investigations, OPSEC-hardened research environments and systems administration. Currently building a secure, on-premise LLM system for confidential document analysis.

EDUCATION

BSc in Information Systems Engineering — Universidad de Alcalá (UAH), Madrid 2022 – 2027
Coursework completed; final thesis (TFG) in progress.
Higher Technician in Telecom & Computer Systems — Salesianos de Atocha, Madrid 2020 – 2022

WORK EXPERIENCE

Cyber Threat Intelligence Intern (OSINT) — Qintegra, Madrid 2022
- OSINT monitoring of the technology sector and production of strategic intelligence reports.
- Created and maintained sock-puppet research personas under strict OPSEC: dedicated hardware, amnesic live OS running from RAM (Tails), prepaid SIMs and segregated networks.
- Initial training in forensically sound data acquisition: bit-for-bit disk duplication with hash-based integrity verification.

IT Technician (Internship) — OpenSpring S.L. / Datatronics, Madrid 2022
- Deployed and administered VMs and servers per client specifications (incl. VoIP); technical support for corporate systems and networks.

TECHNICAL SKILLS

Languages: Python, MATLAB, Java, C++, SQL, Bash, PowerShell, HTML, CSS
Software & Systems: OOP, UML, REST APIs (FastAPI), PostgreSQL, Power BI, Linux, Windows, Docker, Git, virtualization, CCNA networking, Jira
Security: OSINT & cyber threat intelligence, OPSEC (sock puppets, Tails, compartmentalized identities), forensic acquisition fundamentals, network security fundamentals, Tor, cryptography fundamentals
AI: Hybrid RAG, local LLM deployment, AI-assisted development (Claude Code, Codex, GitHub Copilot), evolutionary algorithms, k-NN, SVM, neural networks, k-means, object detection & tracking, NumPy, Numba

CERTIFICATIONS & LANGUAGES

Certifications: Cisco CCNA (Switching, Routing & Wireless), CCNAv7, IT Essentials; Power BI basic course.
Languages: Spanish — native; English — B2 (spoken/written), C1 (reading/listening).

TECHNICAL PROJECTS

Final Thesis (TFG): Secure On-Premise LLM for Confidential Document Analysis 2026 · in progress
- Self-hosted LLM with a hybrid RAG pipeline (BM25 keyword search + dense vector retrieval) so organizations can query internal documents without data ever leaving their infrastructure.
- Security scope: isolated/offline inference, document-level access control at retrieval time, audit logging, and testing against prompt injection and sensitive-data leakage; aligned with ENS / ISO 27001.

Local RAG System for F1 Technical Regulations — GitHub Mar 2026
- Python pipeline (PDF parsing, embeddings, FAISS, locally hosted LLM) for retrieval and question answering over technical documents.

Recipe Social Network (group project, UAH) — GitHub
- Built frontend and backend of a full-stack web app: REST API with FastAPI, PostgreSQL data model and HTML/CSS/JavaScript client.

Tor Network & Dark Web OPSEC Exercise (Security in Distributed Systems, UAH) Feb 2026
- Limits of Tor anonymity against a hardened OPSEC model, with a controlled security-testing proof of concept.

OSINT Digital Footprint Investigation (OSINT Techniques, UAH) 2025/26
- Team exercise reconstructing a subject's routine and location from public sources, with ethical safeguards.

Multi-Object Pedestrian Tracking (group project, Computer Vision, UAH) — GitHub Jan 2026
- MATLAB tracking-by-detection: HOG/ACF and YOLOv4, Kalman filter, Hungarian association; benchmarked on PETS09.

Wind Farm Layout Optimization with CRO (group project, UAH) — GitHub 2025/26
- Python metaheuristic placing up to 50 turbines over an 8760-hour wind series (Jensen wake model); Numba JIT and joblib parallelism (~15x speed-up on the hot routine), beating a random baseline.

Soft-Computing Lab Series (MATLAB & Python, UAH) Sep – Nov 2025
- Genetic algorithms and evolution strategies, k-NN, SVMs, neural networks and k-means image quantization.

Post-Quantum Cryptography Demo (Cryptography, UAH) Apr 2026
- Small Python project experimenting with Kyber-512 (ML-KEM) key encapsulation and comparing it with RSA-2048.